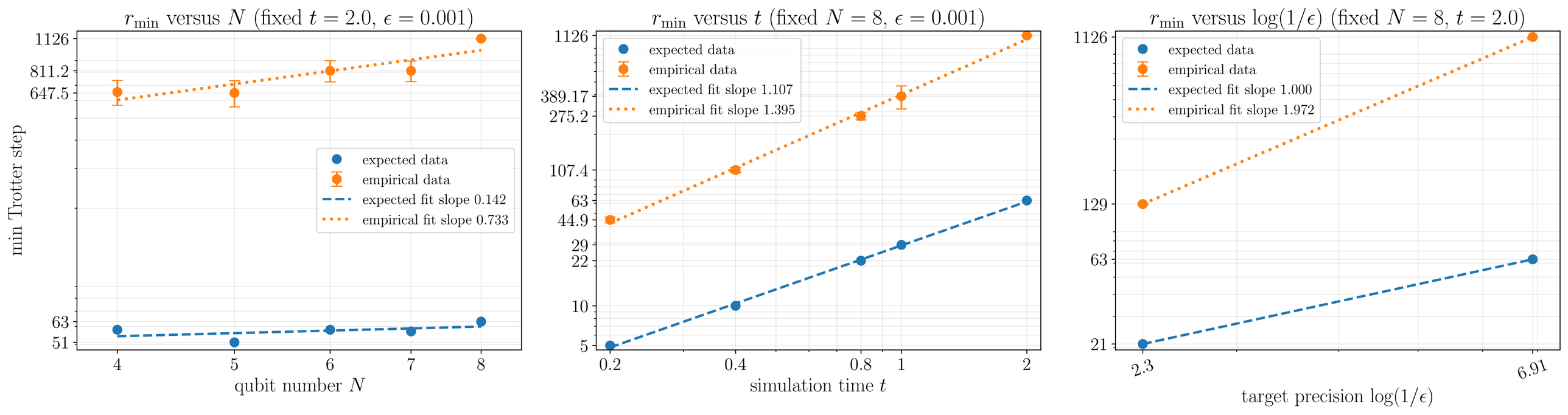

$r_{\min}$ versus N (fixed $t = 2.0$, $\epsilon = 0.001$)
min Trotter step
1126
811.2
647.5
63
51
4
5
6
7
8
qubit number N
expected data
empirical data
expected fit slope 0.142
empirical fit slope 0.733
$r_{\min}$ versus t (fixed $N = 8$, $\epsilon = 0.001$)
1126
389.17
275.2
107.4
63
44.9
29
22
10
5
0.2
0.4
0.8
1
2
simulation time t
expected data
empirical data
expected fit slope 1.107
empirical fit slope 1.395
$r_{\min}$ versus $\log(1/\epsilon)$ (fixed $N = 8$, $t = 2.0$)
1126
129
63
21
2.3
6.91
target precision $\log(1/\epsilon)$
expected data
empirical data
expected fit slope 1.000
empirical fit slope 1.972